Ronald D. Nguyen

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EDUCATION

Northeastern University, College of Engineering **Boston, MA**
Candidate for BS in Computer Engineering and Computer Science, Expected May 2029 *GPA: 3.5*

- Relevant courses: Embedded Design, Circuits & Signals: Biomedical Applications, Fundamentals of Networks

EXPERIENCE

Analogic Corporation **Salem, NH**
Physics Engineer Co-op (6-Month Term) *Jan 2026 – June 2026*

- Developed Linux-based Python scripts using pandas to parse datasets and support production imaging analysis pipelines
- Built a scalable Python CLI tool to validate manual threat-object annotations for lists of any size (typically 80–400 bags), cutting verification time by over 90% (~1 hour to under 5 minutes per list)
- Refactored core C++ codebase into modular components, isolating the data-loading layer from processing logic through a single adapter interface
- Introduced GoogleTest to the C++ codebase with a proof-of-concept suite of 17 unit tests across 6 components, including a custom event listener for per-test pass/fail reporting
- Redesigned a C++ segmentation module around a factory pattern and abstract base classes, enabling algorithm selection via YAML config

PROJECTS

Joystick-Controlled RC Servo **Boston, MA**
FPGA Motor Control (Intel DE1-SoC) *Oct 2026 – Nov 2026*

- Generated an RC PWM signal on a Cyclone V FPGA with LPM counter, comparator, and arithmetic blocks, mapping 0–180° to 0.6–2.4 ms pulses and level-shifting it from 3.3 V to 5 V to drive a servo
- Read an analog joystick through the board's 12-bit ADC and scaled it to servo angle, reusing the PWM design as a modular block
- Measured pulse widths within 0.013 ms of expected on an oscilloscope, and traced an angle mismatch to the ADC's 0–4.096 V input range rather than the assumed 0–5 V

Ultrasonic Sonar Rangefinder **Boston, MA**
FPGA Sensor Interfacing (Intel DE1-SoC) *Oct 2026*

- Interfaced an HC-SR04 ultrasonic sensor with a Cyclone V FPGA over GPIO, generating a 10 Hz trigger and timing echo pulses in 50 MHz clock cycles with LPM counters and comparators
- Converted echo counts to distance with an LPM divider and reused my 8-bit display decoder to show centimeters on 7-segment displays
- Verified readings within 1 cm of ground truth at 10, 20, and 30 cm using oscilloscope measurements, and traced a stuck echo signal to a reversed-power wiring fault

ACTIVITIES

Northeastern Electric Racing **Boston, MA**
Electrical Team Member *Oct 2026 – Present*

- Designed a 2-layer CAN FD transceiver board in Altium Designer for SPARKD onboarding, from hierarchical schematics linked by a signal harness through differential-pair routing and fabrication outputs, passing DRC with zero errors
- Built the CAN interface around a TI TCAN3414 transceiver with a common-mode choke, 60.4 Ω split termination, and TVS ESD protection, powered by a 5 V to 3.3 V LDO

SKILLS

Languages: Python, C++, MATLAB, R
Hardware & Lab: Oscilloscope, function generator, DMM, DC power supply, ADALM 2000 (Scopy), NI USB DAQ, Intel DE1-SoC (Cyclone V FPGA), Arduino
EDA & Simulation: LTspice (DC, transient, FFT), Quartus Prime, Altium Designer
Software & Tools: Git, GoogleTest, CMake, OpenCV, Linux, Windows, AutoCAD, SolidWorks
Concepts: Digital logic design, embedded systems, PWM motor control, ADC and sensor interfacing, inverting and summing op-amp amplifiers, RC circuits, sampling and aliasing, object-oriented design, design patterns, unit testing